Zibo Jiang

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EDUCATION

University of California, San Diego — **San Diego, CA**

M.S. in Electronic Circuits and Systems (EC 78) — *Sep 2026 – Jun 2028*

- **Relevant Coursework**: ECE 284 VLSI Implementation for ML, ECE 260A VLSI Digit Sys Algorithms&Arch, ECE 226 Optimize&Accelerate/Platforms

East China Normal University — **Shanghai, China**

B.Eng. in Electronic Information Science and Technology, GPA: 3.5/4.0 (86/100) Rank: 5/37 — *Sep 2022 – Jul 2026*

- **Relevant Coursework**: Analog Circuit, Digital Circuit, RF Devices and Circuits, Semiconductor Physics, Semiconductor Devices, Information Memory, Artificial Intelligence, Computer Architecture, ASIC

Nanyang Technological University — **Singapore**

Visiting Research Student in Power Electronics, Advisor: Professor Yun Yang — *Jun 2025 – Sep 2025*

University of California, Berkeley — **Berkeley, CA**

Visiting Student, Department of Electrical Engineering and Computer Science — *Jan 2024 – May 2024*

- **Relevant Coursework**: EECS 16A, EE 120 Signals and Systems, EE 105 Microelectronic Devices and Circuits

RESEARCH INTEREST

Hardware Accelerator, Domain-Specific Architectures, Hardware/Software Co-design

RESEARCH EXPERIENCE

Hardware Emulation System for a Memristor based Neural Network Accelerator | ECNU | Research Student

Advisor: Bobo Tian, Professor of ECNU — *Sep 2024 – Jun 2025*

- Designed and implemented a hardware emulation and control system to train and perform inference on neural networks, with the goal of simulating and accelerating the development of a memristor-based compute-in-memory AI chip.
- Developed a customized mixed-signal driver board with 64-channel DACs and a 10-channel transresistance amplifier and ADC readout chain to precisely drive a weight-mapped 64x20 resistor crossbar array.
- Achieved an end-to-end hardware inference accuracy of 95.3% on an image recognition task, demonstrating a minimal 1.7 percentage point gap compared to the 97% software baseline. Click here to view the technical report.

An FPGA Based End-to-End CNN Hardware Accelerator | Tsinghua University | Summer Research Intern

Advisor: Jian Song, Professor of Tsinghua University and IEEE Fellow — *Jun 2024 – Sep 2024*

- Developed an FPGA-based reaction time tester using Vivado, creating a high-precision system capable of measuring user reaction time down to 1μs. Click here to view the project report.
- Designed and implemented a complete end-to-end hardware accelerator on an FPGA for the LeNet-5 CNN model. Utilized Python in a PyTorch environment for CNN model training and used Verilog for the RTL design and verification of parallel computing units, controller, and on-chip memory.

Thermal Management of an Inverter System Based on COMSOL | NTU | Summer Research Intern

Advisor: Yun Yang, Assistant Professor of Nanyang Technological University — *Jun 2025 – Sep 2025*

- Aiming to improve lab's full-bridge Parallel-GaN inverter, which includes 4 GaN HEMTs. Due to the substantial heat generated by the GaN devices, the system could support high-frequency (6.78 MHz) and high-power (2.78 kW) inversion for short durations (0.5s). The goal is to establish a thermal model and optimize the inverter's dissipation performance.
- Built a 1:1 inverter model in COMSOL and simulated the temperature results by setting different GaN heat power dissipation conditions. Compared COMSOL thermal simulation with the experimental thermal imaging results, validating the simulation results were within a reasonable range.
- Through literature review, proposed a cooling strategy utilizing a phase change material (PCM) to absorb the instantaneous Joule heat generated by GaN HEMT devices, extending the inverter's operation time.

A Multimodal Chinese-chess Robot Based on Large Language Models | ECNU | Research Project Leader

Advisor: Shaohua Zuo, Professor of ECNU *Feb 2025 – Jun 2025*

- Developing an intelligent chess-playing robot, integrating camera visual recognition and voice communication hardware modules to achieve natural human-robot interaction.
- Implemented the hardware setup based on ESP32. Developed Python program using Large Language Model API on a host PC. Achieved fluent human-robot conversation.
- Currently training a set of Chinese chess recognition models that will immediately output the chess-playing strategy and robotic arm movement commands by converting movements on the board into LLM prompts.

WORK EXPERIENCE

Shanghai Fosun High Technology (Group) Co., Ltd. **Shanghai, China**

Strategic Development Intern, *Strategy Advancement Department* *Jun 2026 – Aug 2025*

- Developed an LLM-powered web application using Python, Streamlit, and OpenAI APIs to acquire high-quality industry news through natural-language query interpretation, intelligent keyword generation, automated news retrieval, and content extraction. *https://news-data-agent-test.streamlit.app*
- Co-designed and implemented a rule-based scoring framework with separate 100-point relevance and quality assessments, incorporating multi-dimensional scoring, threshold-based filtering, and weighted ranking to identify and prioritize high-quality news articles.

AWARDS

- 2026 Outstanding Graduate of East China Normal University *Jun 2026*
- 2025 Outstanding Student Award at East China Normal University *Dec 2025*
- 2025 Excellent Student Scholarship of East China Normal University *Nov 2025*
- Third Prize, 2024 China Undergraduate Mathematical Contest in Modeling *Dec 2024*
- Third Prize, 2023 Chinese Mathematics Competitions *Dec 2023*
- 2023 Outstanding Student Leadership Award at East China Normal University *Dec 2023*
- Second Prize, 2023 Mathematics Competition, East China Normal University *Jun 2023*

SKILLS

- Programming Languages: Python, C, Verilog
- Software Tools: Vivado, COMSOL, LaTeX (Overleaf)
- Circuit Simulation: MATLAB Simulink, SPICE, ADS
- Hardware Design: Altium Designer, EasyEDA
- Embedded Systems & Hardware Development: FPGA Programming (Xilinx Zynq 7020)